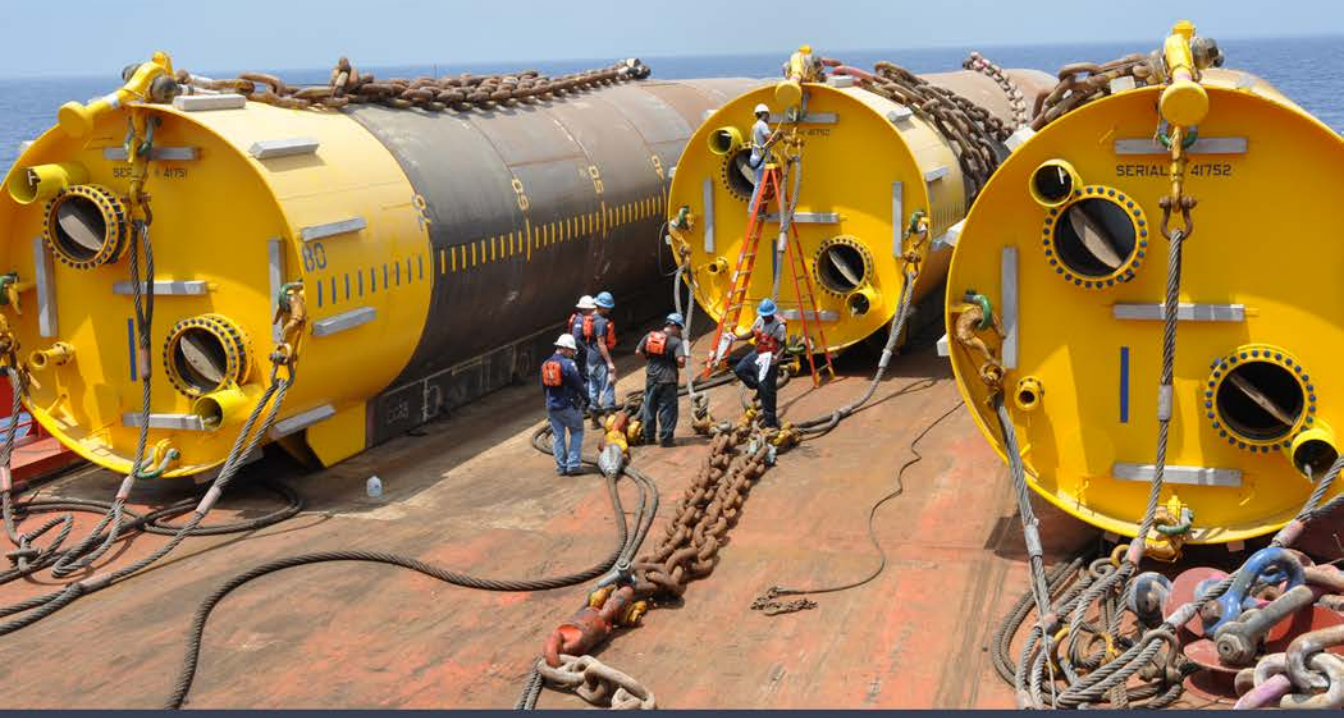

Geomechanics of Marine Anchors

Charles Aubeny

CRC Press
Taylor & Francis Group

Geomechanics of Marine Anchors

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Geomechanics of Marine Anchors

Charles Aubeny

CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

© 2018 by Taylor & Francis Group, LLC
CRC Press is an imprint of Taylor & Francis Group, an Informa business

No claim to original U.S. Government works

Printed on acid-free paper

International Standard Book Number-13: 978-1-4987-2877-5 (Hardback)

This book contains information obtained from authentic and highly regarded sources. Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, please access www.copyright.com (http://www.copyright.com/) or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. CCC is a not-for-profit organization that provides licenses and registration for a variety of users. For organizations that have been granted a photocopy license by the CCC, a separate system of payment has been arranged.

Trademark Notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

Library of Congress Cataloging-in-Publication Data

Names: Aubeny, Charles, author.
Title: Geomechanics of marine anchors / by Charles Aubeny.
Description: Boca Raton, FL : CRC Press, Taylor & Francis Group, [2018] |
Includes bibliographical references and index.
Identifiers: LCCN 2017017150| ISBN 9781498728775 (hardback : alk. paper) |
ISBN 9781498728782 (ebook : alk. paper)
Subjects: LCSH: Offshore structures--Anchorage. | Anchors--Design and construction. | Marine geotechnics.
Classification: LCC TC1665 .A83 2018 | DDC 623.8/62--dc23
LC record available at https://lccn.loc.gov/2017017150

Visit the Taylor & Francis Web site at
http://www.taylorandfrancis.com

and the CRC Press Web site at
http://www.crcpress.com

To My Wife, Ulker,

for her support throughout the development of this book.

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Contents

Preface xiii
Author xv

1 Introduction **1**

1.1 Floating and compliant structures 2
1.1.1 Semisubmersible 2
1.1.2 Tension leg platform 3
1.1.3 Guyed tower 4
1.1.4 Spar 4
1.1.5 Arrays of floating units 4
1.2 Anchor types 6
1.2.1 Suction caissons 7
1.2.2 Piles 8
1.2.3 Drag embedded anchors 10
1.2.4 Suction embedded plate anchors 12
1.2.5 Dynamically installed piles 13
1.2.6 Dynamically installed plate anchors 14
1.2.7 Other direct embedment plate anchors 15
1.2.8 Dead weight anchors 17
1.3 Mechanisms of resistance 18
1.3.1 Soil shearing resistance 18
1.3.2 Gravity 20
1.3.3 Structural resistance 22
1.3.4 Inertia 22
1.4 Loading conditions 24
1.4.1 Sustained loading 24
1.4.2 Cyclic loading 25
1.4.3 Eccentric loading 26
1.4.4 Out-of-plane loading 27
1.5 Anchor performance 28
1.5.1 Installation and load capacity characteristics 28
1.5.2 Efficiency 31
1.5.3 Maintaining embedment 33

1.6 Industry guidelines 34
References 35

2 Soil behavior 39

2.1 Undrained behavior 39
2.1.1 Consolidation stress state 40
2.1.2 Preconsolidation stress 41
2.1.3 Coefficient of earth pressure at rest 42
2.1.4 Advanced laboratory strength tests 44
2.1.5 Normalized behavior 45
2.1.6 Anisotropy 47
2.1.7 Influence of overconsolidation 50
2.1.8 General conditions of consolidation 50
2.1.9 Strain rate effects 51
2.1.10 Remolded strength 53
2.1.11 Thixotropy 53
2.1.12 Sample disturbance 54
2.1.13 Sample reconsolidation 57
2.1.14 In situ tests 58
2.1.15 Undrained stiffness 64
2.2 Consolidation 64
2.2.1 Primary consolidation deformations 65
2.2.2 Secondary compression 67
2.2.3 Coefficient of consolidation and permeability 69
2.2.4 In situ tests 71
2.3 Sand behavior 72
2.3.1 Drained stress–strain-strength behavior 72
2.3.2 Correlations to density and stress level 74
2.3.3 Dilatancy 75
2.3.4 Correlations to CPTU tip resistance 76
2.3.5 Permeability 78
2.4 Cyclic behavior 82
References 87

3 Analytical methods 93

3.1 Plasticity theory 94
3.1.1 The yield criterion 94
3.1.2 The flow rule and internal energy dissipation 96
3.1.3 Generalized stresses and strains 100
3.2 Plastic limit analysis 101
3.2.1 Lower and upper bound theorems 101
3.2.2 Virtual work analysis 101
3.2.3 Method of characteristics 104
3.3 Soil constitutive behavior 106

3.3.1 Linearly elastic–perfectly plastic 107
3.3.2 Modified Camclay 108
3.3.3 Advanced elasto-plastic models 109
3.4 Cavity expansion methods 111
3.4.1 Cylindrical cavity expansion 112
3.4.2 The strain path method 114
3.4.3 Installation disturbance 118
3.5 Finite element methods 119
3.5.1 Overview 120
3.5.2 Coupled poro-elastic analysis 124
3.5.3 Uncoupled consolidation 126
3.5.4 Collapse loads 128
3.5.5 Optimization methods 130
References 132

4 Fundamental studies 135

4.1 Plate anchors 135
4.1.1 Strip anchors 136
4.1.2 Circular disks 138
4.1.3 Eccentric normal loading on rectangular plates 139
4.1.4 Shear–torsion on plates 141
4.1.5 General combined loading 145
4.1.6 Free surface effects 145
4.1.7 Inclined shallow anchors 147
4.2 Cylindrical anchors 150
4.2.1 Lateral loading 150
4.2.2 Axial loading 153
4.2.3 Free surface effects for lateral loading 154
References 159

5 Anchor line mechanics 161

5.1 Soil resistance on anchor line 161
5.1.1 Slack line on seabed 161
5.1.2 Embedded line or chain 162
5.2 Anchor line tension in soil column 163
5.3 Anchor line configuration in soil 165
5.4 Coupling to anchor line in water column 167
5.5 Modification for composite mooring lines 169
5.6 Interaction with a stationary anchor 172
References 173

6 Caisson and pile installation and setup 175

6.1 Suction installation in clay 175
6.1.1 Required underpressure 175
6.1.2 Plug stability 177

6.1.3 *Plug heave* 178
6.1.4 *Practical considerations* 179
6.2 *Suction installation in sand* 179
6.2.1 *Soil resistance for no seepage flow* 180
6.2.2 *Pore water flow* 181
6.2.3 *Estimating soil disturbance effects* 183
6.2.4 *Side resistance and maximum penetration depth* 184
6.2.5 *Required underpressure* 184
6.2.6 *Practical considerations* 186
6.3 *Dynamic installation* 187
6.3.1 *Limit equilibrium* 188
6.3.2 *Analytical solution* 190
6.3.3 *Total energy method* 192
6.4 *Driving installation* 192
6.4.1 *Elastic wave propagation* 194
6.4.2 *Wave equation analysis of piles* 198
6.5 *Setup* 201
6.5.1 *Predictive framework* 201
6.5.2 *Suction caissons* 203
6.5.3 *Dynamically installed piles* 206
6.5.4 *Driven piles* 206
6.6 *Structural integrity* 208
References 210

7 Caisson and pile ultimate capacity **213**

7.1 *Axial capacity in clay* 213
7.1.1 *Driven piles* 213
7.1.2 *Suction caissons* 215
7.1.3 *Jetted piles* 217
7.2 *Axial capacity in sands* 217
7.2.1 *β-method* 218
7.2.2 *CPT methods* 218
7.3 *Horizontal capacity of piles in clay* 221
7.3.1 *Horizontal-moment interaction: $L_f/D > 3$* 221
7.3.2 *Optimal load attachment depth* 224
7.3.3 *Yield locus representation* 225
7.3.4 *Model validation* 227
7.3.5 *Effects of anisotropy* 227
7.4 *Combined loading in clay* 229
7.4.1 *Side axial–lateral interaction* 229
7.4.2 *End axial–moment interaction* 233
7.4.3 *Virtual work solution* 234
7.4.4 *Model validation* 235
7.5 *Long term loading* 236
7.6 *Laterally load piles in sands* 237

7.7 Torsion 239
References 240

8 Elastic effects and soil–pile interaction 243

8.1 Axial load transfer 244
8.1.1 Side resistance 244
8.1.2 Tip resistance 246
8.2 Lateral soil resistance 248
8.2.1 Experimental measurement 249
8.2.2 Application to design 251
8.3 P–y curves for monotonic loading 252
8.3.1 Soft clay 252
8.3.2 Stiff clays 255
8.3.3 Sand 257
8.4 Secant stiffness 258
8.5 True nonlinear behavior 260
8.5.1 Phenomenological description 261
8.5.2 Incremental plasticity models 263
8.6 Soil–pile interaction 265
8.6.1 Axial 265
8.6.2 Lateral 268
8.6.3 Beam-column effects 272
8.7 Short piles and caissons 273
8.7.1 Circumferential distribution of soil resistance 275
8.7.2 Elastic effects and ovalization 277
References 279

9 Drag embedded anchors 281

9.1 Basic description of DEAs 281
9.2 Empirical methods 284
9.2.1 Design guide of NCEL 284
9.2.2 Manufacturer guidelines 285
9.3 Undrained behavior of an embedded plate 286
9.3.1 Load capacity 289
9.3.2 Kinematic response 293
9.3.3 Shank resistance 295
9.4 Embedment trajectory 298
9.4.1 Anchor set 298
9.4.2 Post-set trajectory 300
9.5 Factors affecting DEA performance 302
9.5.1 Anchor properties 302
9.5.2 Soil properties 303
9.5.3 Mooring line properties 303
9.6 Calibration to case history data 305
9.6.1 Joint industry project Gulf of Mexico 307

9.6.2 South Timbalier Block 295 Gulf of Mexico 308
9.6.3 Liuhua 11-1 field 311
9.6.4 Offshore Brazil P-13 311
9.6.5 Offshore Brazil Campos Basin P-27 313
9.7 Vertically loaded anchors 313
9.7.1 Shank release 315
9.7.2 VLA trajectory simulation 316
9.8 Stratified soil profiles 318
9.9 Out-of-plane loading 320
9.9.1 Analysis of thin shank anchors 321
9.9.2 Experimental data 323
9.10 DEAs in sand 326
9.11 Additional considerations 328
References 330

10 Direct embedment plate anchors 333

10.1 Suction embedded plate anchors 333
10.1.1 System of forces on anchor 334
10.1.2 Anchor kinematics 335
10.1.3 Interaction with mooring line 337
10.1.4 Example trajectory calculations 338
10.1.5 Comparison to experimental data 340
10.1.6 Parametric studies 341
10.1.7 Performance of keying flap 342
10.2 Pile driven plate anchors 343
10.2.1 Load capacity of strip anchors in sand 345
10.2.2 Finite length effects 347
10.2.3 Effect of anchor orientation 349
10.2.4 Chain resistance 350
10.2.5 Keying loss of embedment 350
10.3 Dynamically embedded plate anchors 351
10.3.1 Dynamic embedment 353
10.3.2 Keying loss of embedment 355
10.3.3 Load capacity 357
10.4 Helical anchors 358
10.4.1 Axial capacity in clay 359
10.4.2 Axial capacity in sand 362
References 363

Index 367

Preface

Geomechanics of Marine Anchors is intended for students and practitioners with an interest in the design of anchorage for floating structures. Much of the material presented in this book derives from experience with offshore civil infrastructure, which to date has largely been dominated by oil–gas exploration and production. More recently, various renewable energy systems are being proposed and developed that involve floating systems moored to the seabed. A need for efficient, reliable anchors is likely to be particularly important for such systems, owing to the relatively high capital costs of their support structures compared to traditional oil–gas systems. The basic mechanics of anchors used for offshore energy infrastructure also apply to anchors for ships and naval vessels. In fact, a chapter of this book devoted to drag embedment anchors is largely applicable to both offshore civil structures and ships. However, the constraints peculiar to ships, such as a need for rapid deployment and extraction, rule out a number of other types of anchors covered in this book that are often considered for civil infrastructure, such as suction caissons and piles.

Understanding anchor behavior requires a thorough grasp of the medium in which the anchors are embedded. A full chapter of this book is therefore devoted to a review of soil behavior to provide a foundation for later chapters. The presentation on soil behavior presumes that the reader has taken an introductory course in soil mechanics covering soil index properties and classification, seepage, consolidation, elasticity theory, and soil strength principles. Additionally, much of the analytical framework for understanding and predicting anchor performance is cast in terms of ordinary and partial differential equations; therefore, some familiarity with analytical and numerical solutions to differential equations is needed. Overall, the preparation from a typical undergraduate curriculum in civil engineering should provide sufficient background for utilizing this book. Additionally, this book should be considered suitable for specialists in coastal, ocean and mechanical engineering, although some advance preparation on soil mechanics may be needed. Most of the material presented in this book is suited to elective undergraduate and graduate students as well as practicing engineers.

A first objective of this book is to provide the engineer with an organized analytical framework for understanding and predicting anchor behavior. A number of analytical methods appear in the anchor literature for predicting anchor performance, each with their own range of applicability. Chapter 3 presents the available analytical tools, including plastic limit analysis, elastoplastic soil constitutive models, cavity expansion methods and finite element analyses. An effort has been made in the organization of the text to highlight the relative strengths and limitations of the various methods to assist the engineer in selecting the most appropriate suite of analysis methods for a given problem.

In regard to the second objective, a comprehensive presentation of the various anchor systems, the author's first impulse was to devote an entire chapter to each anchor type. Owing to overlapping analysis methodologies and many common features amongst the various

anchors, this was found to generate needless redundancy. Thus, the anchors are grouped into two basic categories comprising cylindrical versus plate geometries. The former include suction caissons, driven tubular piles and most gravity installed (torpedo) piles. The major issues associated with cylindrical anchors—installation, ultimate pullout capacity, and elastic effects—are presented in Chapters 6, 7, and 8, respectively. Five types of anchors are considered in the plate category, drag embedment, suction installed, dynamically installed, pile driven, and helical. Drag embedment anchors largely fall into a category of their own, owing to complex interactions with the chain that govern their installation. Therefore, Chapter 9 is devoted exclusively to drag embedment anchors. The remaining plate anchor types are considered collectively as direct embedment anchors. Chapter 10 systematically addresses three major issues associated with direct embedment anchors: installation, loss of embedment during keying, and ultimate pullout capacity.

In writing a text of this type, a problem always arises in selecting an appropriate depth of coverage for the various topics. In the case of mature technologies, such as driven piles, a vast body of knowledge already appears in the published literature. In these instances, this book presents the basic principles for the sake of completeness, but does not attempt to fully incorporate the wide body of published information already in existence. By contrast, the literature on certain topics, such as a number of the plate anchor systems, is relatively sparse and often not easily accessible to many readers. In these cases, the text is organized largely with a view toward providing comprehensive coverage.

Finally I would like to acknowledge the various collaborations and associations from which I draw my experience with anchors and geomechanics. These include Dr. J.D. Murff and my other colleagues at Texas A&M University, Professor Robert Gilbert and his colleagues at the University of Texas Austin, Professor Mark Randolph and his colleagues at the University of Western Australia, Carl Erbrich and his colleagues at Fugro Advanced Geomechanics, Knut Andersen and his colleagues at the Norwegian Geotechnical Institute, Ed Clukey from the Jukes Group and Arash Zakeri from BP America Inc.

Author

Dr. Charles Aubeny is a professor at Texas A&M University where he is coordinator of the geotechnical group. He is an expert in offshore foundations and serves on the Offshore Site Investigation and Geotechnics Committee for the Houston Branch of the Society of Underwater Technology. He is the recipient of the ASCE Thomas A. Middlebrooks Award and a Fellow of the American Society of Civil Engineers.

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Chapter 1

Introduction

Anchors are a critical component of floating and compliant structures used in coastal and ocean engineering applications. This book presents the application of geomechanics for understanding and predicting anchor performance. Geomechanics provides the analytical framework for characterizing the interaction of the anchor with the soil in which it is embedded. It is central to two aspects of anchor behavior: embedment of the anchor into the seabed to a sufficient depth to ensure that it can fulfill its intended function and the resistance of the anchor to pullout under applied working loads. Other engineering specialties contribute to anchor design, each of which could occupy a volume of its own; these include fatigue and corrosion, mooring line system design in the water column, the dynamics of floating structures, the mechanical hardware for anchor installation, instrumentation for monitoring anchor performance, and the vessels and logistics for transporting and installing anchors. These topics are largely outside of the scope of this book, aside from occasional mention of their impact on geotechnical behavior. Thus, anchor geomechanics is one of a number of specialties contributing to the design of mooring systems, but one of central importance.

The focus of the coverage in this book is largely directed toward anchors for offshore energy infrastructure, historically dominated by oil–gas exploration and production, but with an emerging focus on floating renewable energy systems. Many of the basic principles guiding the analysis of anchors for offshore energy infrastructure apply equally to anchors for ships and naval vessels, and findings from naval research laboratories has often been applied to anchors used for floating energy facilities. However, design constraints can tend to push the development of anchors for these applications in diverging directions. A main driver guiding the design of anchors for offshore energy development is maximizing the pullout load capacity of anchors. Pullout capacity is also a consideration for ship anchors, but other factors constrain their design, such as rapid installation and extraction, and the necessity of avoiding excessively deep embedment to reduce the potential for snagging cables and pipelines. A glance at an anchor load capacity chart will show commonly used anchors for shipping to have an order of magnitude lower capacity than the high-capacity anchors favored by the energy industry. The apparent "inefficiency" of commonly used anchors for ships is driven by constraints that do not normally occur in energy applications.

In regard to organization of this book, this introductory chapter provides an overview of the common types of floating and compliant structures secured by anchors, different categories of anchors, the physical mechanisms by which anchors resist pullout, how the nature of loading affects anchor performance, and criteria for evaluating anchor performance. This introductory chapter contains few quantitative formulations and is intended to provide the reader with an intuitive grasp of these topics; rigorous development of theoretical formulations and analytical methods is saved for later chapters. The remainder of this book comprises three major segments:

- Chapters 2 through 5 cover basic principles relevant to all anchor types, in successive order these include soil mechanics, relevant analytical methods from geomechanics, fundamental solutions that have been applied to all categories of anchors, and anchor line mechanics.
- Chapters 6 through 8 cover cylindrical anchors, with Chapter 6 covering caisson and pile installation, Chapter 7 ultimate load capacity, and Chapter 8 elastic response and soil–pile interaction.
- The final major segment is plate anchors, with Chapter 9 covering drag embedded anchors (DEAs) and Chapter 10 covering various types of directly embedded plate anchors.

1.1 FLOATING AND COMPLIANT STRUCTURES

As development progresses to greater water depths, the cost of fixed structures increases exponentially, driving the use of floating or compliant structures secured to the seabed by anchors. In deep water, it is also more difficult and expensive to hold the natural periods well below the dominant sea state period. The more common of these structures are described below.

1.1.1 Semisubmersible

A semisubmersible (Randall, 1997) is a floating structure comprising large vertical columns supported by large pontoons at the bottom. It floats high in the water while being towed to the site. The pontoons are then flooded to partially submerge the structure, the majority of which is below water. Only the columns are exposed to environmental loading, which contributes to the stability of the structure. A semisubmersible can be moored to the seabed using catenary or taut systems, as illustrated in Figures 1.1 and 1.2. Catenary moorings apply essentially horizontal loads on anchors, while taut moorings are inclined up to approximately 30–35° from horizontal. The mooring lines can comprise chain or wire rope, the latter are lighter but more prone to damage and corrosion. In deep waters, catenary moorings require excessively large offsets, resulting in an increased likelihood of encroaching upon adjacent tracts, crossing mooring lines from adjacent facilities, or crossing undersea pipelines. Hence, for these situations, taut mooring systems provide a more attractive method of mooring floating platforms. In deep water, the weight of the mooring line itself becomes a significant design consideration, a problem which has been addressed through the introduction of lightweight synthetic mooring lines (Vryhof, 2015).

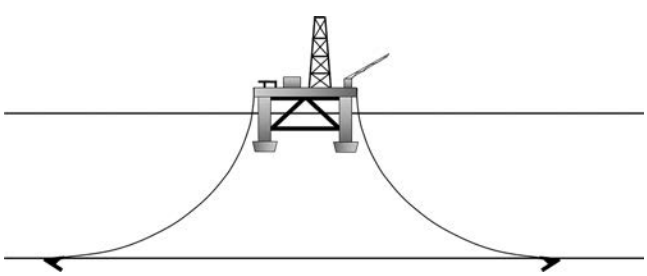

Figure 1.1 Semisubmersible with catenary mooring system.

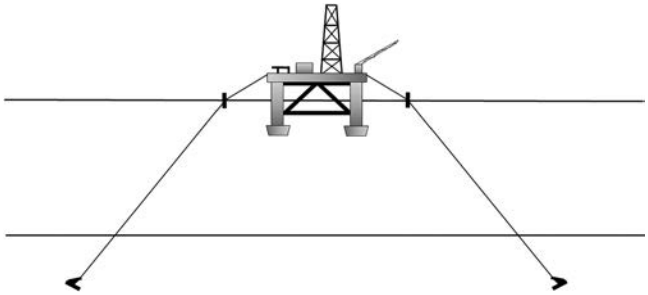

Figure 1.2 Semisubmersible with taut mooring system.

1.1.2 Tension leg platform

A tension leg platform (TLP) consists of a semisubmersible vessel moored by vertical tendons connected to the seafloor, as illustrated in Figure 1.3. The excess buoyancy of the structure—typically on the order of 15%–25% of the platform displacement—maintains the tendons in tension even under the worst storm loading conditions. Conoco constructed the first TLP in the British sector of the North Sea in the early 1980s in approximately 150 m of water. This structure was selected for relatively shallow water as a test of the concept prior to use in deep waters. Since then, several deep water TLPs (>450 m) have been installed. The foundations for TLPs usually consist of large diameter pipe piles that provide resistance to uplift through skin friction, although suction caissons (Section 1.2.1) can be used. Loading on the foundation consists of a tensile mooring force and cyclic loading owing to wave loading on the superstructure. Loading is predominantly vertical with lateral forces estimated at less than 10% of vertical forces. A critical loading condition most likely occurs when a change in mean sea level owing to a storm surge occurs in conjunction with

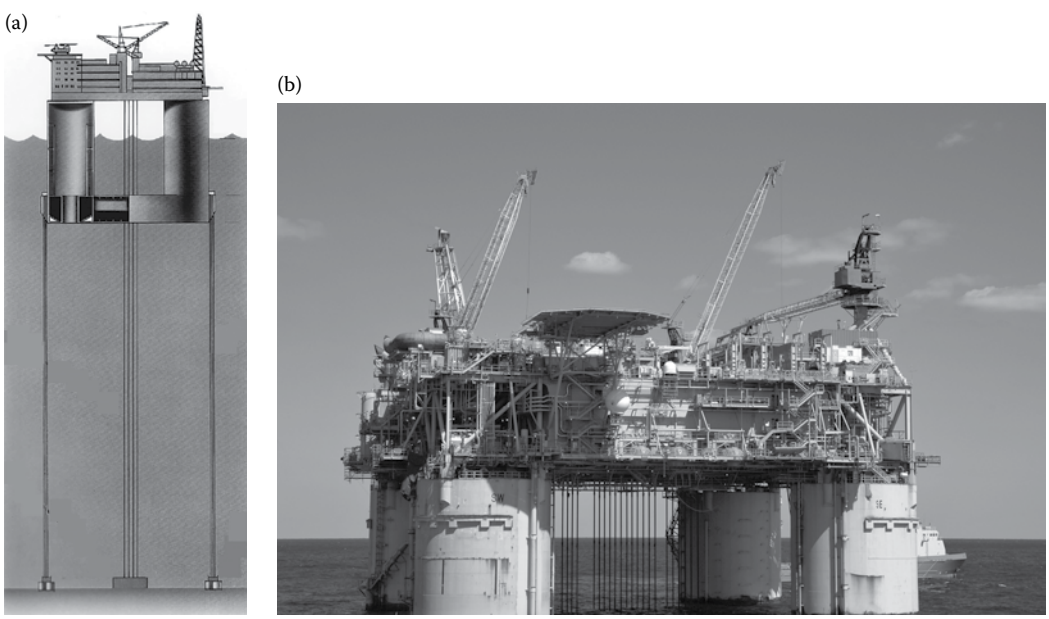

Figure 1.3 Tension leg platform. (a) Schematic and (b) View from surface.

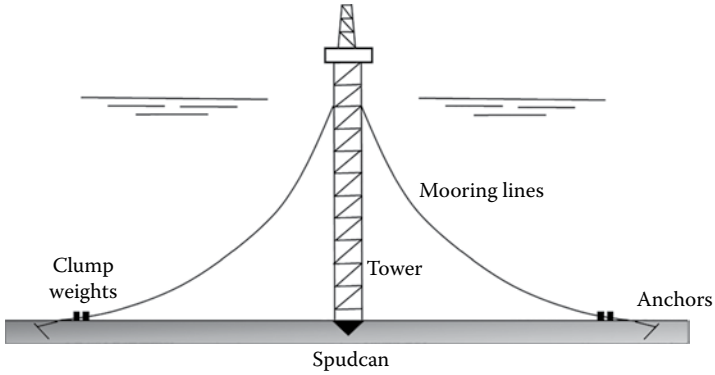

Figure 1.4 Compliant guyed tower.

large wave loading. TLPs can provide a feasible platform alternative to water depths from approximately 450 to at least 1000 m.

1.1.3 Guyed tower

A guyed tower (Figure 1.4) comprises a slender steel truss supported by a compression spudcan foundation and held upright by guy wires secured by anchors. Heavy clump weights are inserted between the anchors and the tower. Under normal conditions, the guy restrains the tower without lifting the clump weights. However, under extreme weather conditions, the clump weights lift from the seabed to provide additional restoring force to resist the wave loading. This concept was first employed for the Lena tower in the Gulf of Mexico in 305 m of water. The tower was secured by 20 wire rope mooring lines and 179 clump weights.

1.1.4 Spar

The Spar platform (Figure 1.5) is an old idea that emerged as a popular deepwater alternative because of various advantages over other solutions. It is essentially a deep-draft, truncated cylinder, with a moonpool that supports a platform by means of excess buoyancy. The hull is a steel cylinder, which encloses soft tanks in the bottom and hard tanks in the top part to provide the required buoyancy. Variable ballast allows adjustments to be made to maintain a constant draft when the topside loads change. The spar can be tied to the sea bottom by vertical tethers but more commonly by catenary or taut mooring lines. In oil-gas applications drilling and production risers run down a centerwell, which shields them from wave and current loading. As a result, the spar is very stable, relatively insensitive to deck loads and easy to transport. All natural periods of moored spars are much longer than that of waves; therefore, their linear dynamic response to wave action is small. There are, however, nonlinear effects, owing to different frequencies, which can cause significant responses in the natural frequencies of the platform, and which should be considered.

1.1.5 Arrays of floating units

Renewable energy can be harvested from sources that include offshore wind, tidal, and wave energy. Much of this energy is located at sites where floating units are likely to be

Figure 1.5 Spar. (a) Schematic and (b) View from surface.

most feasible. For example, over 60% of US offshore wind energy potential is in water depths greater than 60 m where floating offshore wind turbines are more economical than fixed bottom systems. The industry is in a very early stage of development, and the early models for floating offshore wind turbines tend to build on the concepts described above for exploiting offshore hydrocarbon resources. It is uncertain how these will evolve, but a prominent feature of renewable energy development that distinguishes it from oil–gas applications is that floating units are deployed in arrays, rather than isolated single units. This creates an opportunity for increased economy in anchors by attaching more than one mooring line to a single anchor, a concept that is in the very early stages of development (Fontana et al., 2016). Aside from an obvious benefit of cost reductions, multiline attachments provide a promising method for providing redundancy in design at little added cost. In principle, three mooring lines can secure a floating wind turbine, but this leads to a fragile system, as a single damaged mooring line can lead to loss of the unit. Distributing load to additional anchors can also permit smaller, more economical mooring lines. Noting that even temporary moorings for semisubmersibles for oil–gas exploration typically employ 8–12 moorings, multiline moorings can facilitate welcome resilience to renewable energy systems. Conceptually, anchors that possess a vertical axis of symmetry—caissons and piles are discussed later in this chapter—readily lend themselves to multiline moorings, as depicted in Figure 1.6. Nevertheless, plate anchors having a preferential direction of load resistance can still be configured into a multiline arrangement by transmitting mooring line loads through an intermediate attachment ring down to the anchors, although the potential gains in economy will likely be less for plate rather than cylindrical shapes. Since caissons and piles are typically more expensive than plate anchors, multiline mooring systems may

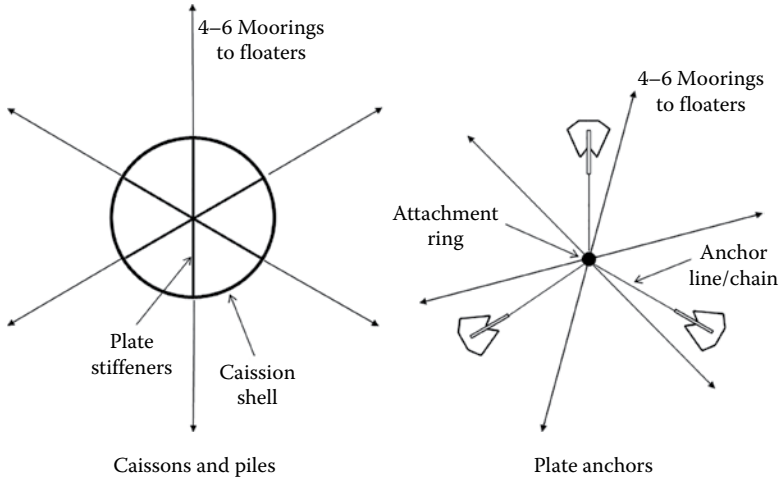

Figure 1.6 Possible multiline attachment configurations.

well level the playing field for selecting the most suitable anchor alternative for floating offshore wind turbines.

1.2 ANCHOR TYPES

This book considers seven types of anchor types. Figure 1.7 shows several example anchors, including driven piles, a suction embedded plate anchor (SEPLA), a suction caisson, a DEA, and a vertically loaded anchor (VLA), alternatively termed a drag embedded plate anchor. Of the seven types considered in this book, three can be considered as relatively mature technologies—suction caissons, piles, and DEAs. Two are relatively recent innovations, but have experienced a sufficient number of field deployments based on which they may reasonably be considered economically viable—dynamically installed piles and SEPLAs. Finally, two anchor types are limited in the presence of commercial offshore deployments. However, ongoing research indicates that they are technically feasible and have promises for economic viability—dynamically embedded plate anchors (DEPLAs) and helical piles. It is emphasized that the term "mature" refers not so much to their level of technological development as to the scale of their deployments to date. For example, piles and DEAs are among the oldest of anchor alternatives and have experienced widespread deployment, but there is plenty of room for innovation and improvement of predictive models of performance for these anchors.

The various anchors have advantages and limitations in regard to the soil types and loading conditions for which they are most suitable, which will be the focus of most of the discussion that follows. Although this book focuses primarily on geotechnical aspects of anchor performance, some mention in cost differences is essential to guide the discussion. In this regard, DEAs may be considered as the least expensive of the mature technologies, piles most expensive, with suction caissons somewhere in between. In terms of scale, a factor of five between least and most expensive is not unrealistic; however, it is emphasized that costs are highly sensitive to project-specific factors such as equipment availability and vessel costs. Moreover, cost comparisons should be considered in light of differences in reliability amongst the various anchor types.

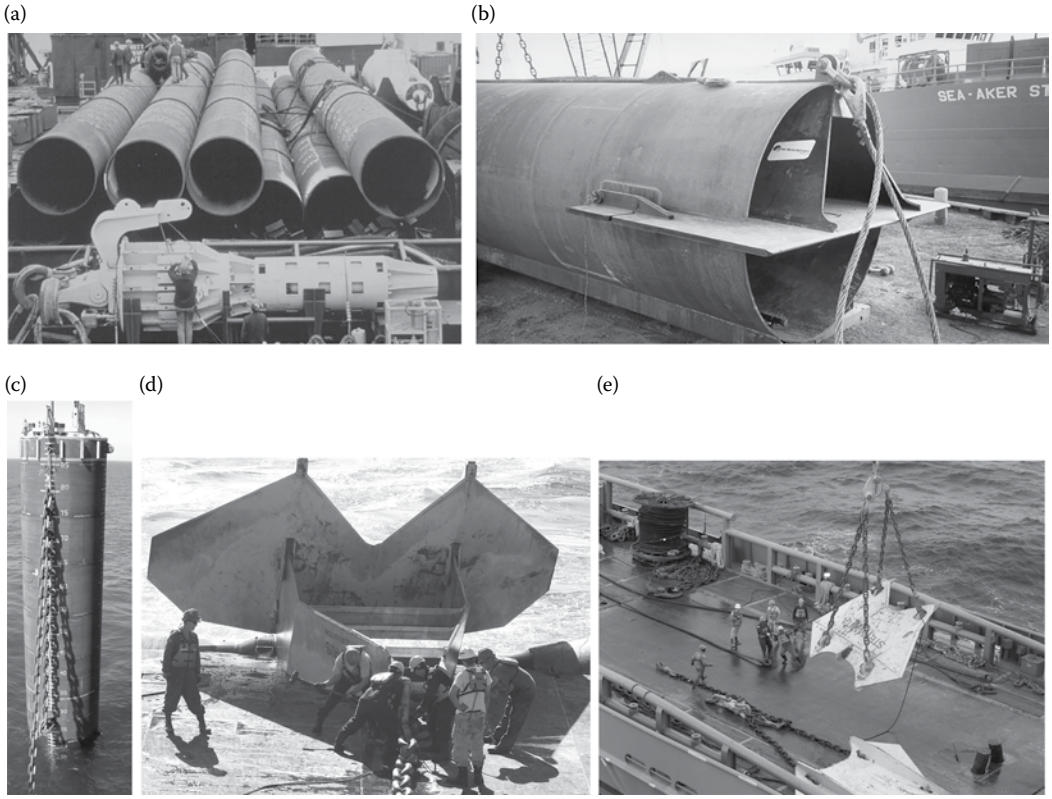

Figure 1.7 Example anchors. (a) Driven piles, (b) suction embedded plate anchor (SEPLA, courtesy of Intermoor), (c) suction caisson, courtesy of E. Clukey (d) drag embedment anchor (DEA, courtesy of Delmar), and (e) VLA, vertically loaded anchor, courtesy of Delmar.

1.2.1 Suction caissons

Suction caissons (Andersen et al., 2005) are large cylindrical anchors that are installed partially by self-weight penetration, with penetration to full installation depth accomplished by the application of "suction," which is actually a differential pressure induced by pumping through a valve in the top cap (Figure 1.8). Suction caissons have advantages over piles in that they avoid the requirement for heavy underwater hammers. Moreover, they do not require auxiliary platforms such as jack-up rigs to support the installation operation. Suction installation is feasible in a range of soil profiles, including soft clays, stiff clays, and sand. Suction installation in clays is a matter of applying sufficient underpressure to overcome the various sources of soil resistance that are described in detail in Chapter 6. The maximum underpressure is limited by considerations of stability of the plug of soil inside the caisson—excessive underpressure can cause the soil plug to heave into the interior of the caisson. Installation in sands also relies on underpressure to advance the caisson, but the underpressure also induces upward seepage flow inside the caisson which dramatically reduces the resistance to penetration. Caisson aspect ratio (ratio of length to diameter) is largely dictated by considerations for safely installing the caisson. Optimal aspect ratios are typically in the range of 1.5–3 in stiff clays, less than 1.5 for dense sands, and greater than 5 for soft clays. Although suction caissons are versatile from the standpoint that they can be installed in a wide range of soil profiles, heterogeneous soil profiles are problematic.

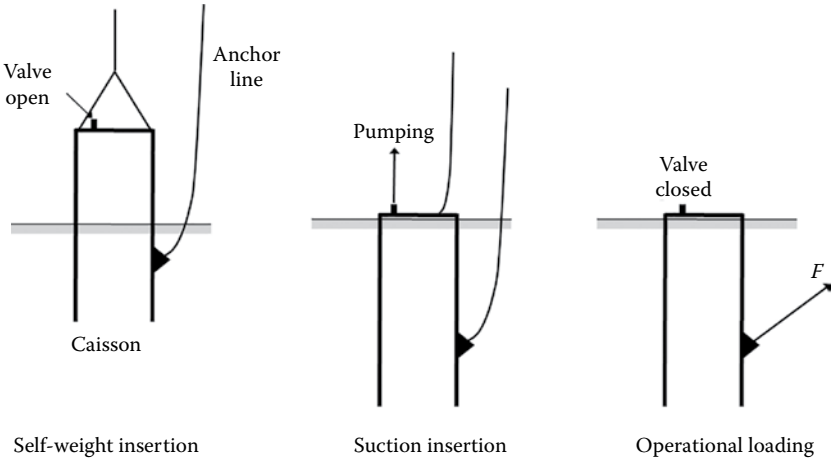

Figure 1.8 Suction caisson installation.

For example, interbedded clays can interrupt the upward seepage that is essential for reducing resistance to penetrate in sands. Recent developments in suction installation (Tjelta, 2015) show that suction installation is possible in heterogeneous soil profiles by cycling or jetting, but the process is more complex—and requires considerably more expertise—than in the more benign homogeneous soil profiles. Proof testing to verify load capacity is not normally required for suction caissons (Andersen et al., 2005), since their positioning is well controlled, their design is supported by numerical studies calibrated to model tests, and their design is based on the data of site-specific soils.

In general, suction caissons can resist both horizontal and vertical loads, and are therefore suitable anchors for either catenary or taut mooring systems. For reasons to be discussed subsequently, suction caissons experience substantial reductions in load capacity when subjected to sustained vertical loads. For this reason, they are not particularly attractive for anchorages involving this type of loading, such as TLPs. Horizontal load capacity of suction caissons is sensitive to the depth at which the load is applied; therefore, an optimal load attachment (pad eye) depth that produces maximum capacity should be selected. Methods for estimating the optimal load attachment depth as well as predicting ultimate load capacity are presented in Chapter 7. Suction caissons are normally designed with relatively thin walls; typical diameter-to-wall-thickness ratios are in the range of 125–160, in contrast to 10–40 for piles. Various internal plate stiffeners and ring stiffeners are typically required for preventing buckling failure during installation (Section 6.6) and structural failure during operational loading. Wall-thickness selection and stiffener design is within the province of structural engineers, but geotechnical engineers are typically called for providing soil reaction pressures to support the structural analyses, a topic addressed in Chapter 8.

1.2.2 Piles

In spite of the many attractive features of suction caissons, piles remain a viable anchor alternative in many instances. Piles (Figure 1.9) can be installed in a wide range of geologic profiles, including profiles containing calcarenite layers and rock (Vijayvergiya et al., 1977) that would normally not be considered feasible for suction caissons. Additionally, heterogeneous soil profiles that, as described above, present a challenge for suction caisson installation, are

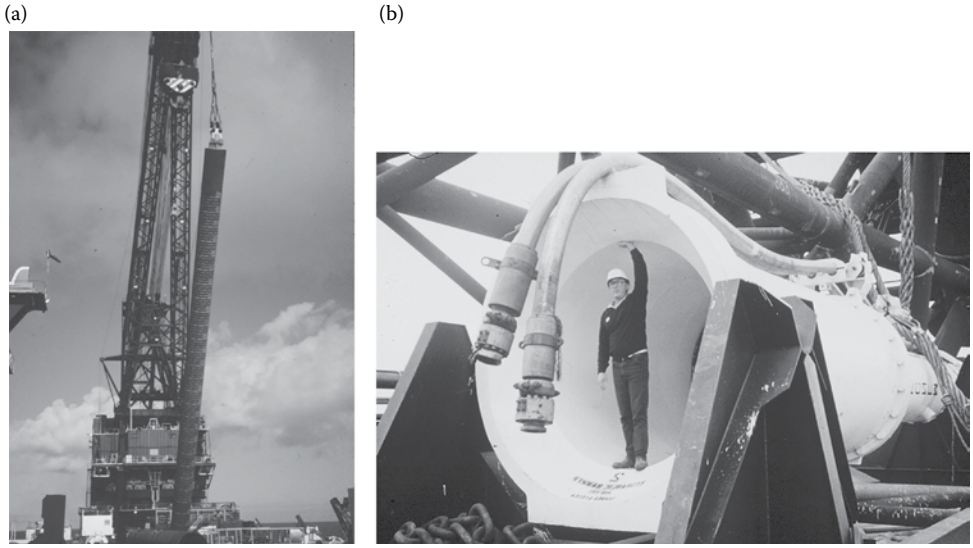

Figure 1.9 Driven pile. (a) Tubular pile and (b) Hammer.

not particularly problematic for driven pile installation. In regard to load resistance capabilities, the shortcomings of suction caissons in resisting pure vertical load were noted earlier. For this reason, piles remain the anchor of choice for TLP systems, which involve nearly pure vertical loading on anchors. Piles can also resist horizontal loads. Short piles—piles with length-to-diameter ratios less than 6—experience negligible bending when subjected to horizontal and moment loading. Thus, their mechanical behavior under applied loads is similar to suction caissons, and the limit analysis methods presented in Chapter 7 for caissons also apply to short piles. Longer piles behave as axially–laterally loaded slender structural members, and prevailing practice models piles as beam columns supported by "soil springs" representing the continuum in which they are embedded. While elastic response can be potentially significant for any anchor type, slender piles are the only anchor type considered in this book for which elastic behavior can be said to truly dominate their design. Chapter 8 presents methods for evaluating the elastic response of piles to mooring line loads.

While a wide variety of pile types, materials, and methods of installation have evolved in general geotechnical practice, much of offshore pile practice uses driven open-ended tubular piles, with diameters frequently in excess of 2 m. Onshore methods for driving piles include impact hammers powered by steam, air, or diesel; and, in cohesionless soils, vibratory hammers that advance the pile by liquefying the soil while applying a downward load. Historically, onshore technology was adapted to offshore pile installations using a "follower" that was attached to the pile to transmit the stress wave from the water surface to the pile. This approach was technically feasible to water depths up to approximately 300 m. In the mid-1970s, development at greater water depths spurred the development of underwater hydraulic hammers. Various developments in this technology are described by Cox and Christie (1976), Jansz (1977), Heerema (1980), Aurora (1984), and van Luipen (1987). Wave equation analyses are used for estimating driving stresses, hammer selection, and penetration rates. They are also used in a reverse sense to back-analyze driving records to estimate pile load capacity.

An alternative to hammer driven installation is jetting, which is commonly used for conductor installation in oil–gas applications (Zakeri et al., 2014). Jetting involves extraction

of soil through the interior of the pile to reduce resistance to penetration. This mode of penetration is in contrast to conventional pile driving by which the volume of the penetrating pile is accommodated by displacing the soil. Although jetted piles will likely have less than half the axial capacity of driven piles, they have some appeal for offshore renewable energy applications. Objectionable acoustic emissions associated with pile driving are relatively limited for oil–gas projects involving some 10–20 piles; however, environmental concerns become more serious in renewable energy projects potentially involving hundreds or even thousands of piles. As a much quieter installation method, jetting may well be a favorable alternative to driving from the standpoint of acoustic emissions. Since much of pile design is empirical, largely based on load tests on driven piles, adoption of jetted piles would likely require a recalibration of much of the current design methodology. A second alternative to mitigating acoustic emissions is a "bubble curtain"—a ring of bubbles streaming upward from the seabed to interrupt sound wave propagation (Love and Arndt, 1948). However, the effectiveness of bubble curtains may be limited owing to sound transmission through the seafloor (Lee et al., 2011). Further, difficulties arise in attenuation of low frequencies, which requires large stable freely rising bubbles. Lee et al. (2011) investigate the use of tethered latex balloons to overcome this difficulty.

1.2.3 Drag embedded anchors

A DEA is a bearing plate (termed the "fluke") inserted into the seabed by dragging with wire rope or chain (Figure 1.10). The fluke is attached to the anchor line by a "shank" comprising one or more plates. Self-embedment of the anchor is achieved by controlling the line of action of the anchor line force—by setting the fluke–shank angle—such that "failure" occurs roughly parallel to the fluke so that the anchor will dive downward when dragged. DEAs are normally designed for several possible fluke–shank angle settings according to soil type. In stiff clays and sands, the fluke–shank angle is typically on the order of 30°, while 50° is typical for soft clays. DEAs are less expensive to install relative to caissons and driven piles and, as will be discussed subsequently in more detail, plate anchors are also generally more efficient than caissons and piles in terms of the ratio of load capacity to anchor weight. However, DEAs do not have the advantage enjoyed by caissons and piles in terms of their ability to be precisely positioned. Further, the load capacity of the anchor depends on the depth of anchor penetration, which cannot be predicted with a high degree of uncertainty. Nevertheless, the uncertainty in DEA load capacity can be mitigated substantially by

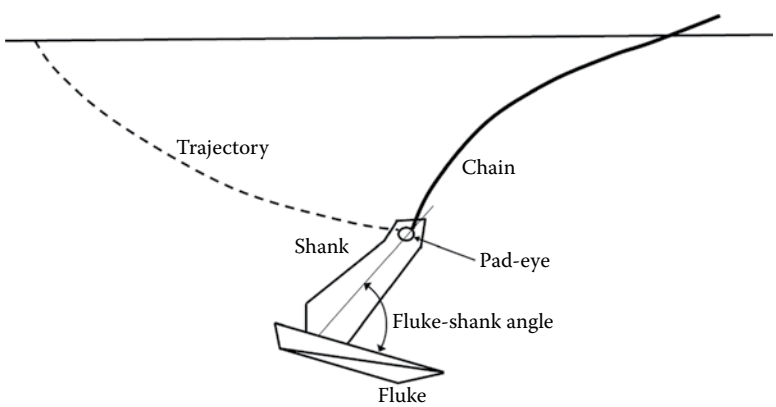

Figure 1.10 Drag embedded anchors.

proof load testing of anchors following installation, which is usually required for permanent installations. It should be noted that obtaining a vessel capable of proof testing is not a trivial matter, as the load capacity of some anchors may exceed the available bollard pull of many anchor handling vessels (AHVs). Winch-driven tensioners may alleviate this obstacle in cases where another anchor is available to provide the necessary reaction force (Vryhof Anchors, 2015).

DEAs in sands and stiff clays experience minimal penetration into the seabed, typically less than 1–2 fluke lengths. Consequently, these anchors have minimal vertical load capacity and their suitability is restricted to catenary systems. Effectively, the anchor provides resistance to horizontal loads, while the dead weight of the anchor chain resists vertical loads. The extensive lengths of anchor chain required for this arrangement may easily exceed the costs of the anchors; thus, in comparing anchor alternatives, it is important to compare the cost of the entire mooring system, rather than considering only the cost of the anchors. Although caissons and piles are more expensive than DEAs, they typically have vertical load capacities suitable for taut moorings. The reduced mooring line costs for such systems may well offset the differences in anchor costs.

In contrast to the case of sands and stiff clays, DEAs in soft clays will penetrate to substantial depths, on the order of tens of meters in some cases. Since soft clay profiles typically exhibit increasing strength with depth, increased penetration leads to increased DEA load capacity, including a substantial capability for resisting vertical loads. Thus, DEAs in soft clay can provide suitable anchorage for both catenary and taut mooring systems. To enhance the pullout resistance of DEAs in soft clays, so-called VLAs were developed. A VLA is installed in the same manner as a DEA, but it features a releasable shank that can be opened after drag installation. Most VLA designs employ a shear pin that ruptures when the mooring line force exceeds a certain level, although more recently some mechanical release mechanisms have been developed. Two types of shank have evolved for VLAs: a rigid bar, and a bridle arrangement comprised of chains (Figure 1.11). Since chain mooring lines tend to inhibit anchor penetration, VLAs are usually used in conjunction with wire rope. As noted earlier, the fluke–shank angle of a DEA in soft clay is typically set at approximately 50° to promote diving of the anchor during drag embedment, but maximum pullout capacity of the anchor is developed when the shank is normal to the fluke. Thus, increasing the fluke–shank angle at the end of drag installation provides added pullout capacity. A down side of increasing the fluke–shank angle is that it leads to a failure mechanism involving upward movement of the anchor when the anchor is overloaded. This behavior is undesirable because, as the anchor migrates upward into weaker soil, it rapidly loses load capacity

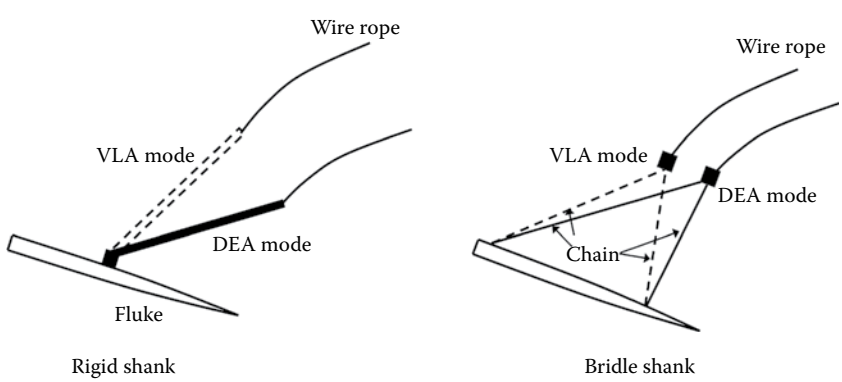

Figure 1.11 Vertically loaded anchors.

leading to a brittle failure mode. To reduce the potential for a brittle failure mode, most VLAs now feature a "near-normal" orientation for the released shank, with the opened shank oriented to slightly less than 90° from the fluke—78° is the largest angle known to the author. At this shank orientation, additional load capacity is achieved when the shank is released. However, if overloaded, the anchor retains a tendency to dive downward. While the near-normal concept definitely leads to reduced tendency for upward movement, the downward motion of the anchor is conditional on the orientation of the anchor line and fluke at the time of failure. Therefore, it is not advisable to simply assume *a priori* that an overloaded VLA will continue to dive downward. Chapter 9 provides further details on this topic.

As both DEA and VLA load capacity depend on embedment depth, the load capacity prediction requires two key calculations: (1) prediction of the anchor trajectory and (2) prediction of load capacity for the depth of embedment achieved. The former calculation depends on the interaction between the anchor chain or wire line. Historically, predictions of drag anchor performance were based on model tests at the prototype and laboratory scale (e.g., Naval Civil Engineering Laboratory, 1987). More recent trends have experienced the application of analytical methods, including the introduction of limit equilibrium analyses (Stewart, 1992; Neubecker and Randolph, 1995a; Dahlberg, 1998), finite element studies and plastic limit analyses of soil–anchor interactions (Rowe and Davis, 1982a; O'Neill et al., 2003; Murff et al., 2005), and analytical formulations for mooring line behavior and interaction between the mooring line and the anchor (Vivatrat et al., 1982; Neubecker and Randolph, 1995b; Aubeny and Chi, 2010).

1.2.4 Suction embedded plate anchors

The SEPLA conceived and developed by Dove et al. (1998) and Wilde et al. (2001) is a plate anchor installed by attaching it to the tip of a suction caisson (Figure 1.12). The suction caisson is installed in the conventional manner, but retracted by overpressure to leave the plate behind. The vertically oriented plate is then "keyed" to turn into the direction of the applied mooring line load. Some loss of embedment occurs during the keying process, with a concomitant reduction in load capacity (Gaudin et al., 2006); consequently, considerable research attention has been devoted to analyzing the loss of embedment. Song et al. (2009) indicate that, for typical loading conditions, keying results in a loss of embedment of

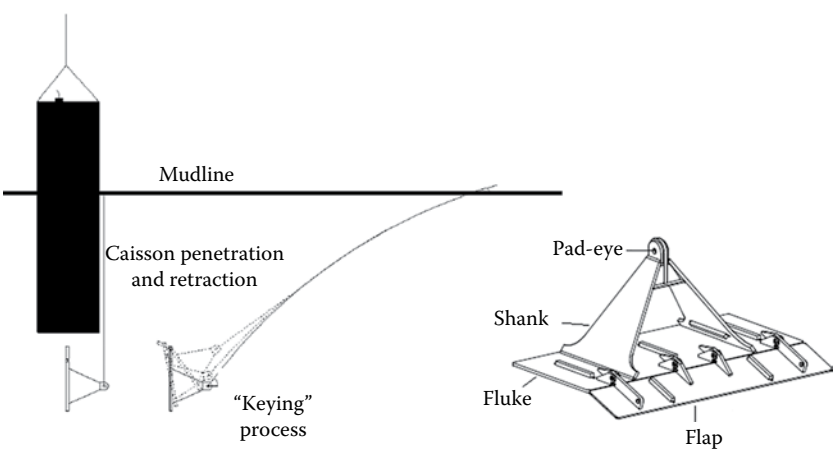

Figure 1.12 Suction embedded plate anchor.

approximately 0.4–0.6 times the height of the anchor. Some SEPLA design concepts incorporate a flap designed to minimize the loss of embedment during keying.

SEPLAs have the appeal that they combine the precise vertical and horizontal positioning of a suction caisson with the lightness and efficiency of a plate anchor. Deployment is largely limited to soft clay soil profiles. Since the anchor is deeply embedded, it has capabilities for resisting vertical loads and is a suitable anchorage alternative for taut mooring systems. Installation time is lengthy—on the order of 20% greater than for a conventional suction caisson installation (typically 1 day)—since both suction installation and overpressure retraction are required. Accordingly, installation costs are greater than that for drag embedded plate anchors, and the benefit of precise positioning of the plate is obtained at the price of higher installation costs. Moreover, transport costs should be factored into any cost comparisons. Owing to their size, suction caissons require large transport vessels, which can be a major cost driver. By contrast, SEPLA deployments require the transport of only single suction caisson in conjunction with a large number of compact plate anchors that can easily be arranged on the deck of a transport vessel. Overall, for soft clay soil profiles where capabilities for resisting vertical/inclined loads and precise anchor positioning are a significant concern, SEPLAs can be considered a competitive alternative.

1.2.5 Dynamically installed piles

Dynamically installed piles, sometimes termed "torpedo piles" penetrate into the seabed by the kinetic energy obtained from free fall through the water column and their own self-weight. This type of anchor (Figure 1.13a) typically is rocket shaped or cylindrical with a nose cone, with up to four stabilizing fins at the top of the pile. The Delmar Omni-Max anchor is somewhat of an exception in that it comprises three plates oriented at 120° from one another rather than a cylindrical body (Figure 1.13b). In this sense, it may be classified as a plate anchor. However, since its mode of installation more closely follows other gravity installed piles, it is included in this discussion. The piles are typically 12–15 m long and 0.8–1.2 m in diameter, with dry weights in the range of 500–1000 kN (Randolph et al., 2011). Reported free fall heights are as high as 170 m (Ehlers et al., 2004), but are more commonly in the range of 50–75 m (Zimmermann et al., 2009; Lieng et al., 2010). Prediction of impact velocity as the pile falls through the water column is a critical part of the overall prediction of penetration depth into the seabed and load capacity; guidelines on the analysis and

(a) (b)

Figure 1.13 Dynamically installed piles.

selection of a drag coefficient are given by Hasanloo et al. (2012). Impact velocities are up to 30 m/s (Randolph et al., 2011). Penetration in soft clays is typically 2–3 pile lengths.

Dynamically installed piles offer the advantage of quick, economical installation. They can be installed with a single AHV and require minimal mechanical equipment—in contrast to the heavy underwater hammers required for piles or pumps for suction caissons. Ehlers et al. (2004) report successful deployments in soil profiles that include normally consolidated clay, overconsolidated clay, sand and calcareous deposits. Considering one comparative study of transportability, Zimmermann (2009) reported that the AHV under consideration could transport in a single trip either 4 suction caissons, 8 torpedo anchors, or 12 VLAs. Thus, economies in transport can be realized for torpedo piles, but not perhaps as much as for plate anchors. Since they embed relatively deeply, dynamically installed piles can resist both horizontal and vertical loads and are thus suitable for both catenary and taut mooring systems.

Prediction of load capacity requires accurate prediction of penetration depth which, in turn, requires accurate prediction of the impact velocity as well as accurate accounting of the effects of rate-dependent soil shearing resistance and inertial resistance. Rate-dependent soil shearing resistance is a particular challenge, since the strain rates occurring during dynamic penetration lie well above the range for which most experimental evaluations of soil strength are conducted. Performance can be enhanced through judicious positioning of the pad eye. Shelton (2007) and Zimmermann et al. (2009) report that placing the pad eye of the Omni-Max anchor near the pile tip promotes diving behavior when the anchor is overloaded.

1.2.6 Dynamically installed plate anchors

A DEPLA is a plate anchor that is installed by free falling through the water column in a manner similar to dynamically installed piles. This anchor concept has the advantage of the lightness and efficiency of a plate anchor combined with the low installation cost of a dynamically installed pile. Two approaches have evolved in the development of dynamic installation of plate anchors. The first approach, developed by O'Loughlin et al. (2014), consists of attaching the plate anchor to a detachable torpedo pile—a "follower"—as depicted in Figure 1.14. The follower can be detached from the plate anchor by triggering a shear pin in a manner similar to the shank release mechanisms for VLAs. The anchor itself comprises two crossed circular plates to form a cruciform configuration. After installation, the pile is retracted and the plate is keyed into position in a manner similar to the keying process for SEPLAs. DEPLA installation using a follower is essentially a dynamic pile installation; therefore, the analytical framework for modeling installation of the two anchor types is similar. The DEPLA concept has been validated through centrifuge model tests, numerical and analytical studies, and reduced-scale field trials, based on which a rigorous design procedure has been developed by O'Loughlin et al. (2016). While this anchor alternative is novel, the technology is promising. O'Loughlin et al. (2015) estimate that the reductions in fabrication costs alone for dynamically installed plate anchors can lead to cost savings of 70%–80%, and this does not allow for the potentially far greater cost savings associated with reduced AHV costs. The studies to date have focused on DEPLA installations in soft clay soil profiles.

The second DEPLA concept involves direct embedment of the plate anchor from free fall without the use of a follower to enhance penetration. A "Flying Wing" anchor based on this concept is currently under investigation. Preliminary single gravity model tests indicate the potential for deployment in both soft clays and sands. As with the other DEPLA approach, the plate is oriented vertically following installation and must then be keyed into position.

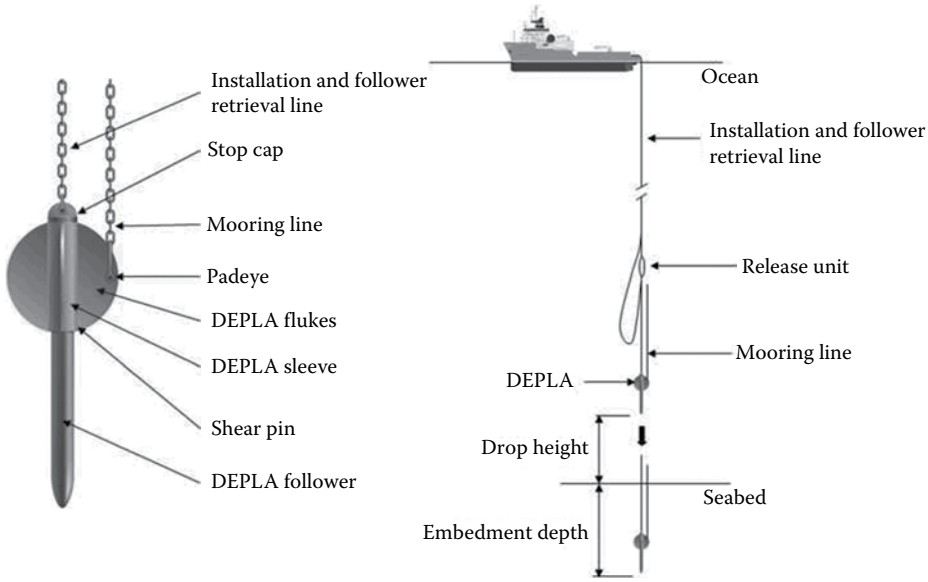

Figure 1.14 Dynamically embedded plate anchor.

This anchor concept is a novel approach. Therefore, continuous research is required for establishing fundamental questions in regard to penetration depth, loss of embedment owing to keying, and bearing resistance of the anchor.

1.2.7 Other direct embedment plate anchors

In addition to the SEPLA described earlier, other forms of directly embedded plate anchors include helical anchors and driven or jetted plate anchors. These types of anchor can be deeply embedded in clays or sands, providing the capability for developing vertical load capacity in sands. Helical anchors have received relatively little attention for offshore floating structures in the energy industry, although they have long been in use for anchoring vessels and submarine pipelines. A helical anchor (Figure 1.15) comprises one or more helical plates having a carefully controlled pitch welded to a central shaft. The central shaft functions to transmit torque during installation and transmit axial loads to the helical plates. Installation torque correlates to anchor load capacity, thereby providing a basis for confirming design load capacity predictions. In addition to their versatility from the standpoint of soil type, helical anchors have an advantage of relatively rapid, inexpensive installation. Small diameter helical anchors can be economically installed by remotely operated underwater vessels (ROVs). Helical anchors have the capability to resist lateral as well as axial loads (Prasad and Narasimha Rao, 1996), but the helices appear to enhance the lateral efficiency compared to a single shaft pile. Current models are relatively small compared to many marine anchors. Elkasabgy and El Naggar (2015) characterize a 0.601-m helix as being "large diameter," which is not inaccurate for land-based applications, would be considered relatively small in marine applications. However, if their value can be shown for marine applications, it is likely that they can be scaled upward to achieve larger capacities.

For estimating load capacity of single-helix anchors in clay, early studies include the investigation of Vesic (1971), Meyerhof and Adams (1968), Meyerhof (1973), and Das (1978, 1980), from which semiempirical design equations were developed. More recently, rigorous

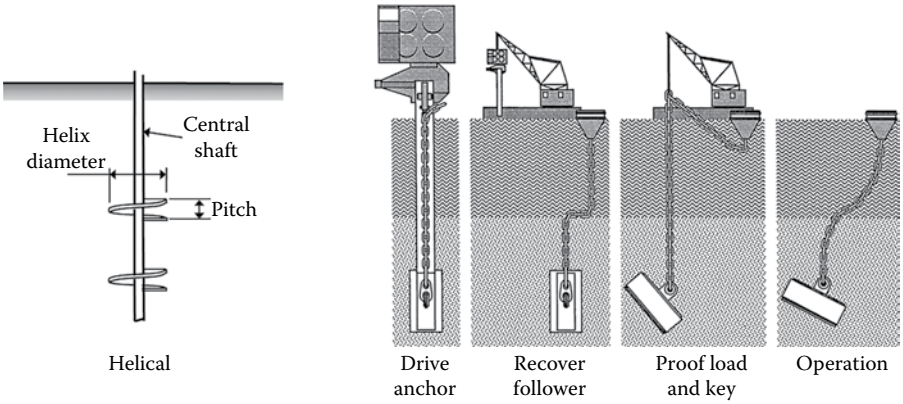

Figure 1.15 Direct embedment plate anchors.

theoretical solutions based on plasticity theory were developed for flat (i.e., the effect of pitch is neglected) circular plates in cohesive soil (Martin and Randolph, 2001; Merifield et al., 2003), which are described in Chapter 4. For multihelix anchors, interference and overlapping stress zones between adjacent plates complicate the analysis. Merifield (2011) presents theoretical solutions and proposed design charts for multihelix anchors in clays having up to three plates. A somewhat similar trajectory of development evolved for sands, with early empirical methods for estimating load followed by successively more rigorous limit equilibrium, cavity expansion, plasticity, and finite element solutions. Giampa et al. (2017) evaluate the accuracy of these capacity prediction methods through comparisons to intermediate-scale anchor tests. A finite element plane strain solution by Rowe and Davis (1982b), adjusted for three dimensional effects, proved to be the method that compared most favorably to measurements.

The solutions described above apply to monotonic loading, with no consideration of cyclic load capacity or upward displacements under cyclic loading. Model tests on cyclically loaded single-plate helical anchors in medium dense sand by Newgard et al. (2015) indicate that the anchor experiences upward ratcheting movements, which lead to significant reduction in anchor capacity under cyclic loading relative to monotonic load capacity. As discussed by Newgard et al. (2015), improved performance in regard to cumulative cyclic displacements may be possible through anchor redesign or using additional anchor plates. Additionally, rather than considering measures purely associated with anchor design, operation and maintenance measures may be employed. For example, damaged moorings may be reinstalled following storm loading, or retorqueing to recover embedment loss may be possible.

Aside from the helical concept for direct embedment, plate anchors can be installed by attaching a plate to a pile tip, which is driven or jetted into the seabed (Forrest et al., 1995; NAVFAC, 2011). The pile is subsequently extracted for reuse. The installation largely employs existing pile installation technology. A subcategory of this type of anchor is the "umbrella" anchor having vertically oriented plates during installation that rotate to a horizontal position during keying. Similar to helical anchors, these anchors enjoy the advantage of precise positioning. Moreover, they can be installed in a broad range of soil deposits including soft clays, stiff clays, dense sands, glaciated soils, and corals. They can be keyed to orient them to the anticipated direction of loading; hence, they are capable of resisting inclined and vertical loads. The low skin friction associated with jetting installation,

normally considered a liability for piles, is actually beneficial, as it facilitates extraction of the pile at the end of plate installation.

1.2.8 Dead weight anchors

In certain situations, such as rock seabeds, penetration into the seabed is not possible, rendering the anchor systems described above impractical. In these cases, consideration may be given to dead weight anchors. Reinforced concrete or scrap steel may be used for this purpose, but are likely to be costly. As a less expensive alternative, Flory et al. (2016) describe a dead weight anchor comprising a ballast-filled flexible bag. The ballast can be any pourable materials, such as sand, drilling mud, or steel pellets. The bags are then grouped within a containment net, as shown in Figure 1.16.

Vertical capacity is simply the submerged weight of the anchor. Load capacity for the system is relatively small—on the order 5 tonnes for the Flory et al. system. In regard to horizontal load capacity, Flory et al. (2016) report a friction coefficient of 0.45 for this

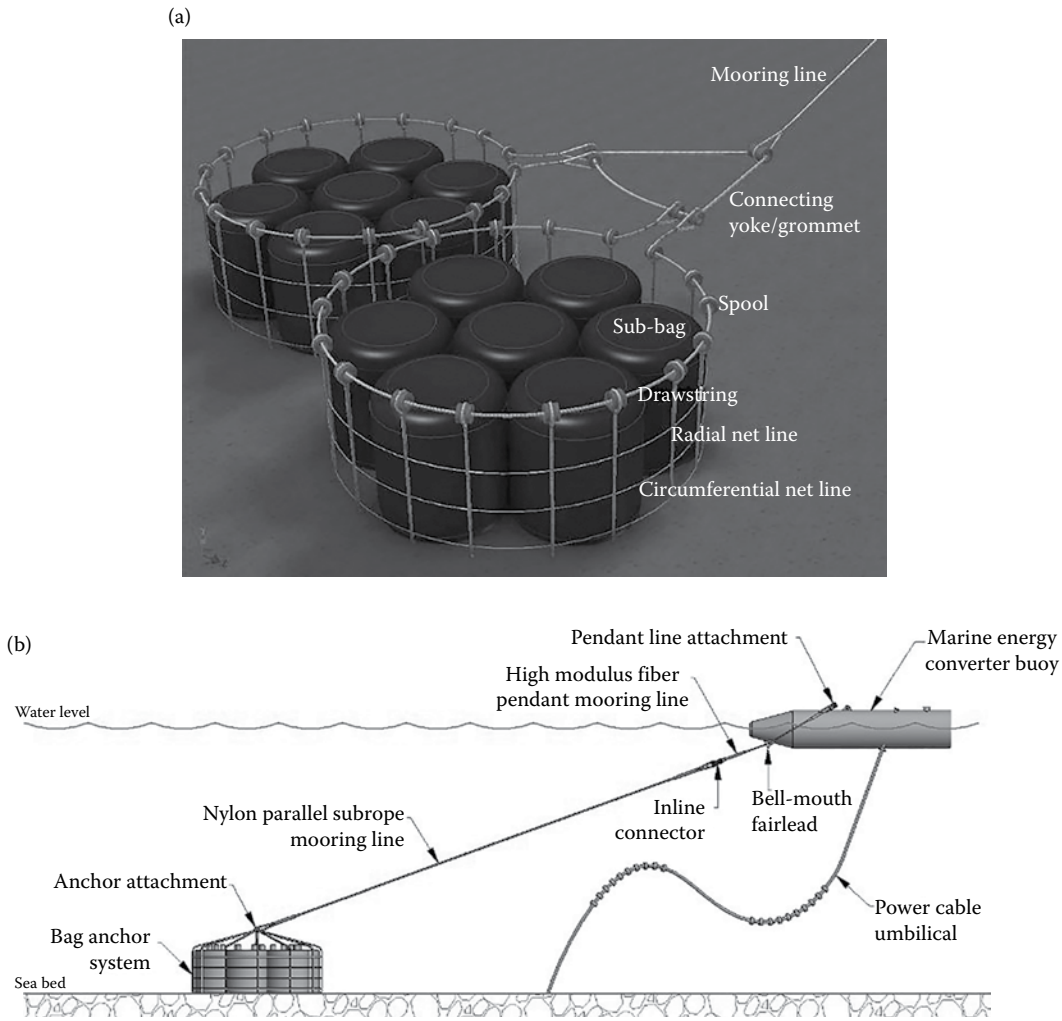

Figure 1.16 Dead weight anchors. (a) Anchor bag system and (b) Attachment to buoy.

system. These capacities are a small fraction of what can be achieved by pile/caissons and plate anchors. Nevertheless, in applications where the load demand is small and seabed conditions are difficult, a simple dead weight system may merit consideration.

1.3 MECHANISMS OF RESISTANCE

The physical mechanisms of resisting forces acting on an anchor should be understood to analyze both anchor installation and anchor pullout capacity. Methods for evaluating resistance comprise much of the content of this book. These methods can be easily misapplied if an intuitive understanding of the basic mechanics is lacking. The following discussion focuses on intuitive concepts, with more rigorous treatment left for later chapters.

1.3.1 Soil shearing resistance

Soil shearing resistance is often the primary source of pullout capacity for an anchor. Shearing resistance may be in the form of frictional resistance acting on the surface of the anchor or bearing resistance within the soil mass in the vicinity of the anchor. Figure 1.17 illustrates the issues associated with calculating skin resistance for the case of piles and caissons in clay. Firstly, pile installation disturbance weakens the soil through the process of remolding as well as the generation of large excess pore pressures that reduce the effective stress, $\sigma'_{rii} < \sigma'_{h0}$. Over time "setup" occurs, primarily owing to the process of consolidation; i.e., the dissipation of excess pore pressures induced during installation. Thixotropy—the time-dependent increase in remolded clay at constant water content—can also contribute the strength gain during the early stages of consolidation (Jeanjean, 2006). Subsequent axial loading induces changes in total stress $\Delta\sigma_{rr}$ as well as generation of additional excess pore pressures Δu. Depending on the rate of axial loading, dissipation of excess pore pressures can occur, such that partially drained or fully drained conditions prevail. Although the process is often described as "side friction," actual slippage can occur either at the soil–pile interface or entirely within the soil adjacent to the pile. Strain-softening behavior at the soil–pile interface can lead to progressive failure as described by Randolph (2003) and presented in detail in Chapter 8. Analytical methods that have been developed for simulating pile installation disturbance and setup, discussed in Chapter 3, provide useful insights into

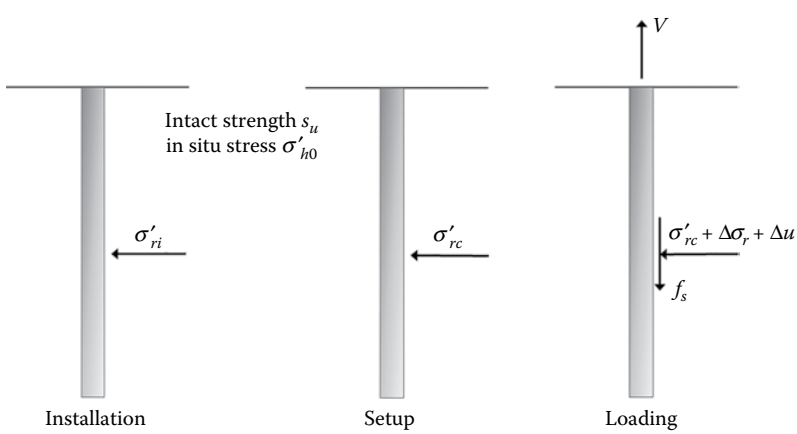

Figure 1.17 Skin resistance on piles in clay.

the factors that can affect side resistance. However, current prevalent practice employs the "α -method," where measured skin resistance is simply correlated to the intact soil undrained shear strength, $f_s = \alpha\ s_u$. The parameter α characterizes the net effect of a number of complex processes and, not unexpectedly, wide variations in back-calculated α occur for similar site conditions. Values of α can vary from nearly unity in normally consolidated clays to less than one-half in over-consolidated clays. Side resistance calculations for plate anchors generally parallel the α-method approach for piles, although the database of supporting physical measurements is less extensive.

Side resistance for piles in sand broadly parallels the discussion above in regard to clays, but differs in a number of details. Installation disturbs the soil but excess pore pressures rapidly dissipate. Therefore, a setup process dominated by dissipation of excess pore pressures does not occur. Nevertheless, time-dependent setup occurs for piles in sands, although the physical mechanisms are poorly understood. Observations of time-dependent increase in side resistance in sands date back to the 1970s; an excellent review of previous studies as well as presentation of recent studies of the effects of pile aging are given by Gavin et al. (2015). Although the mechanisms of aging are poorly understood, field measurements indicate that time-dependent increases in radial effective stresses and interface dilative stresses contribute to increased shaft capacity. Reported setup times are on the order of 1 year. As described in Chapter 7, shaft resistance is predicted based on correlations to *in situ* effective stress or, more recently, correlations to cone penetration test tip resistance.

Bearing resistance involves mobilization of soil shearing resistance outside of the immediate area of the soil–anchor interface, as depicted in Figure 1.18. As the resisting soil mass is largely undisturbed, the analytical estimates of bearing resistance using plasticity theory are often possible, at least for simple loading conditions. Chapter 3 describes plasticity theory and Chapter 4 presents some fundamental solutions that have wide applicability to anchor problems. Bearing resistance in clays is usually expressed as a multiple of the soil undrained shear strength, for example, $q_{ult} = N_c\ s_u$. Not surprisingly given that it involves mobilization of shearing resistance well outside the boundaries of the anchor, bearing resistance is much more effective than frictional. Chapters 3 and 4 present bearing factors for various shapes that vary from approximately $N_c = 7.5$–13, in contrast to the multiplier on soil strength for

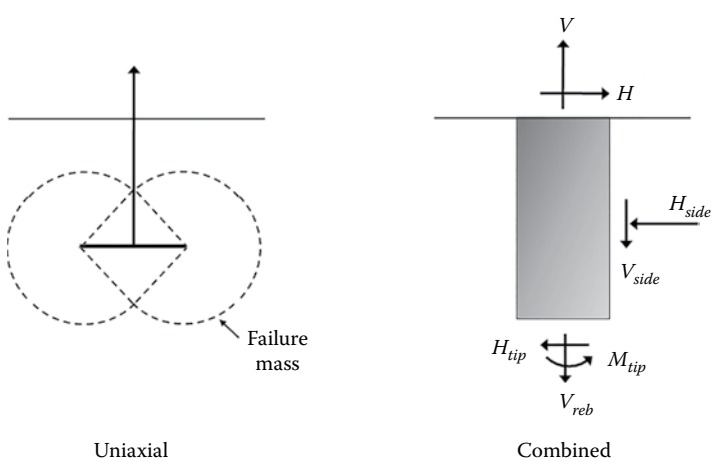

Figure 1.18 Bearing resistance and combined loading.

side resistance, $\alpha < 1$. Thus, a plate anchor that functions by mobilizing bearing resistance can have the pullout capacity as a friction pile many times its size.

Despite the importance of plasticity theory, most analytical solutions apply to "uniaxial" load conditions such as pure translation relative to a given axis of the anchor or pure rotation. Many practical situations arise that deviate from simple uniaxial loading. For example, the sketch in the right-hand side of Figure 1.18 shows a caisson subjected to combined horizontal and vertical loading as would occur in a taut mooring system. In this case, the soil resistance on the side of the caisson is a combination of horizontal bearing resistance and axial side friction. The resistance at the tip is an even more complex combination of vertical reverse end bearing, horizontal sliding and moment resistance. With some unusual exceptions—Chapter 4 gives an analytical solution for combined shear-torsion loading on a plate—analytical solutions are not available for these types of load combinations. In these instances, recourse is generally made to numerical finite element or finite difference solutions to map out a "yield locus," defining the load combination at which yield occurs.

1.3.2 Gravity

The contribution of soil self-weight to anchor load capacity is a common source of confusion, because the contribution only occurs under certain circumstances. We can first consider the case of a horizontally embedded plate anchor shown in Figure 1.19. The sketch on the left considers the case in which no gap develops between the soil and the underside of the plate. This condition occurs when gravity forces are sufficiently large relative to soil shearing resistance such that the soil flows in to fill any gap that may tend to form beneath the plate. Quantitative assessment of this condition is discussed in Chapter 3. The absence of gapping is beneficial from the standpoint of mobilized bearing resistance, since shearing resistance is mobilized both above and below the plate. However, it turns out that soil weight makes no contribution to anchor capacity in this case. When described in terms of virtual work, the positive work performed by the upward-moving soil above and below the plate is exactly balanced by the negative work performed by the soil flowing down around the sides, resulting in zero net contribution of soil weight to anchor capacity. The contrasting condition of gap formation is shown on the right-hand sketch in Figure 1.19. In this case, the plate must overcome the weight of the overlying soil in addition to the soil shearing resistance. In general, gapping is detrimental to anchor capacity, since the added contribution of soil self-weight typically does not offset the loss of bearing resistance beneath the plate. The soil self-weight contribution is nevertheless not negligible and should be considered when gapping is presumed to occur.

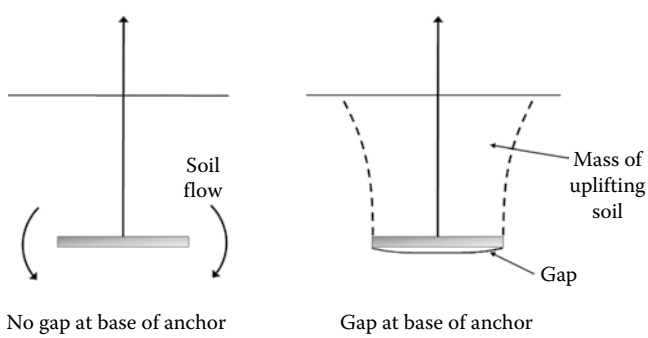

Figure 1.19 Soil self-weight contribution to plate anchor capacity.

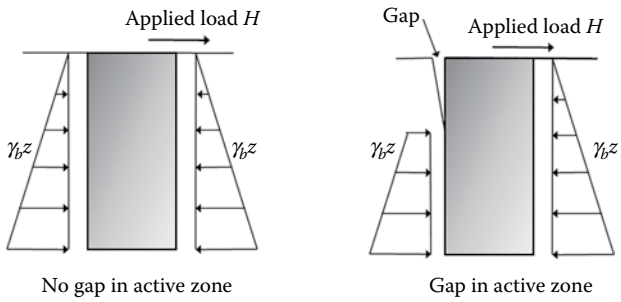

Figure 1.20 Soil self-weight contribution to horizontal load capacity of piles and caissons.

The effect of soil self-weight and gapping on horizontal load capacity of piles and caissons is similar to that described above for plates. As shown in the sketch on the left-hand side of Figure 1.20, in the absence of gapping, the self-weight stress from the soil acts equally on the active and passive sides of a horizontally loaded pile. Thus, the net contribution of soil self-weight to horizontal load capacity is zero. However, in the presence of gapping, the horizontal stresses in the active zone drop to zero as the soil separates from the pile wall. As is the case for the plate anchor, gapping results in a loss of shearing resistance in the region in the vicinity of the gap; however, this loss is partially offset by an added contribution from soil self-weight. Detailed analysis of this effect is given in Chapter 4.

Soil self-weight can also contribute to vertical pullout capacity of suction caissons, although not for the potential failure modes commonly considered in suction caisson design. Figure 1.21 shows the potential modes of vertical failure in caissons. The first assumes that reverse end bearing mobilizes at the base of the caisson. In this case, the virtual work argument stated above for plate anchors applies: the positive virtual work performed by soil inside the caisson is exactly offset by the negative virtual work of the downward moving soil outside the caisson; therefore, soil weight makes no contribution to pullout capacity. The second possible failure mode is a pure pullout failure, which is typically associated with sustained loading where suction cannot mobilize to provide reverse end bearing resistance. By definition, no soil movement occurs for this failure mechanism, so soil weight also does

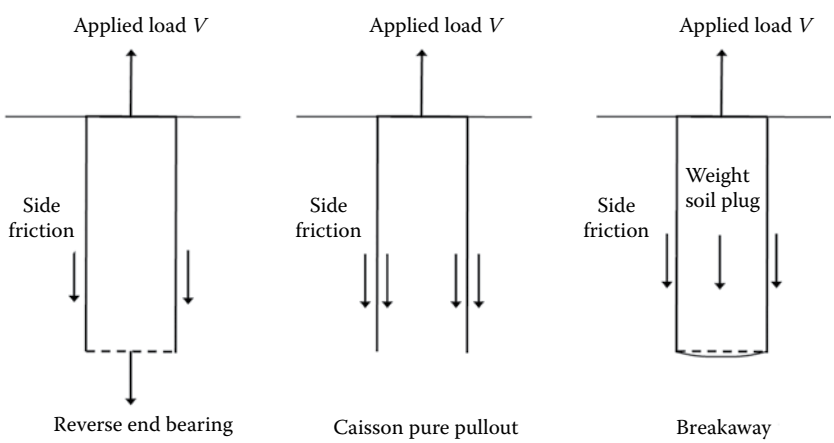

Figure 1.21 Possible vertical failure modes for suction caissons.

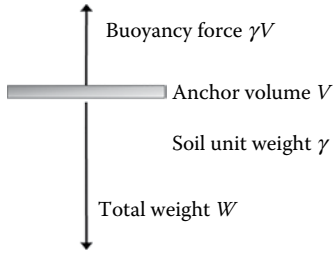

Figure 1.22 Buoyant weight of anchor.

not contribute to pullout capacity. The final possible mode involves loss of reverse end bearing resistance with the side friction on the interior of the caisson being sufficient to cause the caisson and interior soil plug to move as a unit. In this case, the weight of the soil plug clearly contributes to soil resistance.

The weight of the anchor itself will also contribute to vertical load capacity, although the contribution is typically relatively minor. When evaluating the contribution of anchor weight, the buoyant weight of the anchor should be used. In this case, the buoyant weight is the weight of the anchor minus the weight of the material that is displaced by the anchor (Figure 1.22). When the anchor is fully embedded in the soil, the calculation should be based on the total unit weight of the soil. For a steel anchor (specific gravity = 7.8), the buoyancy effect reduces the total anchor weight by approximately 20%–25% for typical soil densities. If part of the anchor protrudes above the mudline—for example, the top cap of a suction caisson—the weight of the protruding portion should be reduced by the displaced volume of water in this portion of the anchor.

1.3.3 Structural resistance

Anchor load capacity is largely synonymous with "geotechnical" capacity; that is, capacity derived from soil shearing resistance and, to a lesser extent, the weight of the soil. The most significant exception to this generalization is laterally loaded flexible piles (Figure 1.23). As will be discussed in Chapter 8, elastic deformation, specifically pile curvature, is so great that the pile fails structurally in bending well before the ultimate shearing resistance from the soil is fully mobilized; thus, the load capacity of the pile itself becomes a primary factor controlling ultimate load capacity. Analytical treatment of this behavior may follow the elastic analysis methods for piles described in Chapter 8. Alternatively, the plastic limit methods discussed in Chapter 3 can also be applied to flexible piles, provided that a plastic hinge is placed in the pile to account for structural yield (Murff and Hamilton, 1993; Chen et al., 2016). Suction caissons typically have aspect ratios less than 6, which is too short for significant beam bending action to occur; thus, they are typically treated as rigid bodies for which geotechnical resistance controls the ultimate load capacity.

1.3.4 Inertia

Rapidly applied loads can mobilize two forms of additional soil resistance: increased shearing resistance owing to the rate dependence of soil undrained shearing resistance, and resistance associated with particle acceleration within the moving soil mass. The former is a soil strength issue, which will be discussed separately. The latter is termed "inertial resistance," which is discussed here. Short duration impact loads for which inertial resistance may be

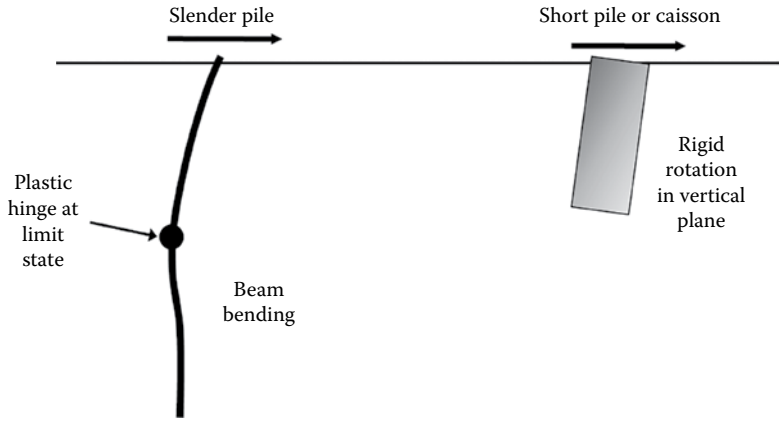

Figure 1.23 Flexible and rigid piles.

significant can arise from sources such as a ship impacting a mooring, various installation and construction operations, and blast loading. Hermann (1981) states that in situations where the load duration (Figure 1.24) is less than 0.02 seconds, the inertial effects may have to be considered. For load durations less than 0.0035 seconds, the Naval Civil Engineering Laboratory (NCEL) guidelines show a tripling of anchor resistance owing to inertial effects. This guidance applies to plate anchors used by naval vessels, it will not necessarily apply to all dimensions and anchor types considered in this book. Nevertheless, it does convey some sense of the time scale of impulse loading for which inertial effects may be significant.

Aside from short-duration impulse loading, significant inertial resistance can arise during continuous penetration processes involving sufficiently high velocities, such as those occurring during dynamic pile installation. True (1974) includes drag resistance (Figure 1.24) in the summation of forces resisting penetration during dynamic pile installation and characterizes the resistance in terms of a drag force similar to that from classical fluid mechanics. Subsequent studies on dynamic installation by Kim et al. (2015) found that inertial forces

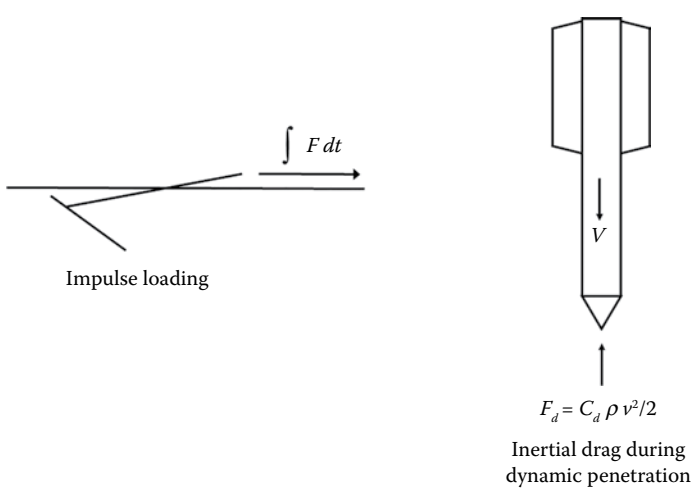

Figure 1.24 Inertial forces on anchors.